

Quiz7 - Results



Attempt 1 of 1

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Attempt Score **10 / 10 - 100 %**

Overall Grade (Highest Attempt) **10 / 10 - 100 %**

Question 1

1 / 1 point

Which of the following best describes the intuition for inverse document frequency (IDF)? (Single Answer)

- ☐ If a document contains a term N times, it is N times more relevant to that term than a document that contains it 1 time.
- ☐ If a document contains a term N times, it is less than N times more relevant to that term than a document that contains it 1 time.
- ☐ If a word occurs in very few documents, that word is obscure and its importance should be reduced.
- ✓ ☒ If a word occurs in many different documents, that word is common and its importance should be reduced.

Question 2

1 / 1 point

A Loss-of-function (knockout) experiment removes a component to test if it is *necessary* for a behavior.

- ✓ ☒ True
- ☐ False

Question 3

1 / 1 point

Why is training a neural relevance classifier to evaluate every single (query, document) pair considered a "bad approach" to Information Retrieval?
(Single Answer)

- ☐ It cannot utilize sophisticated neural models like BERT to predict relevance.
- ✓ ☒ Handling a query would require $O(\text{documents})$ time, making it too slow for large collections.
- ☐ It requires formatting the output as a sparse vector, which consumes too much memory.
- ☐ It relies entirely on inverted index structures, which degrade accuracy.

Question 4

1 / 1 point

When applying a causal steering vector to alter a model's behavior during generation, at which layers does this intervention typically work best?
(Single Answer)

- ☐ The very first embedding layers.
- ✓ ☒ The middle layers.
- ☐ The final unembedding layers.
- ☐ It works equally well across all layers regardless of depth.

Question 5

1 / 1 point

How does "clamping" differ from standard "steering" interventions? (Single Answer)

- ☐ Steering modifies the model's parameters permanently, whereas clamping does not.
- ☐ Clamping is used to isolate Attention heads, whereas steering is strictly used for MLPs.
- ✓ ☒ Steering applies a uniform shift (addition) to a concept vector, whereas clamping sets the strength of an SAE feature to a fixed value.

- ☐ Clamping adds a vector during the backward pass, whereas steering adds it during the forward pass.

Question 6

1 / 1 point

What is a key difference between training a "linear probe" and "fine-tuning" a model? (Single Answer)

- ☐ Fine-tuning freezes all model parameters, while probing backpropagates through all parameters.
- ✓ ☒ Probing freezes all model parameters and only trains a small number of additional parameters for a new classifier.
- ☐ Probing requires unsupervised data, while fine-tuning exclusively requires supervised data.
- ☐ Probing is used strictly for transformer attention heads, while fine-tuning is used for MLP layers.

Question 7

1 / 1 point

What are the key limitations of relying solely on a Language Model's internal knowledge without an external retrieval system? (Select all that apply)

- ✓ ☒ The model's internal knowledge is outdated because it is constrained by a "knowledge cutoff date" from when it was trained.
- ✓ ☒ Models lack access to internal company documents and very long documents.
- ✓ ☐ Models are unable to answer simple, frequently discussed historical facts, such as when the Declaration of Independence was signed.
- ✓ ☒ Models struggle to provide accurate answers to questions regarding long-tail knowledge.

Question 8

1 / 1 point

When using an inverted index for sparse vector querying, which of the following optimizations can be applied to find the top K documents efficiently without exhaustively checking every posting list? (Select all that apply)

- ✓ ☒ Only look at posting lists for terms with a high Inverse Document Frequency (IDF) score.
- ✓ ☒ Only consider documents that appear on multiple posting lists (e.g., using list intersection).
- ✓ ☐ Only look at posting lists for terms with a low Inverse Document Frequency (IDF)
- ✓ ☐ Process the inverted index through a Dense Passage Retriever (DPR) before applying TF-IDF.

Question 9

1 / 1 point

Which of the following describe valid ways to improve the efficiency of dense retrieval? (Select all that apply)

- ✓ ☒ To partition the vector space into K cells by running K-means clustering on the original document vectors.
- ✓ ☐ Store dense document vectors generated by DPR in an inverted index.
- ✓ ☒ Break residual vectors into smaller vectors and quantize them to cluster centroids.
- ✓ ☐ To automatically expand a text query with similar words using a thesaurus.

Question 10

1 / 1 point

Which of the following are accurate descriptions of the "Logit Lens" technique? (Select all that apply)

- ✓ ☐ Logit lens is used in the process of training Sparse Autoencoders (SAEs).
- ✓ ☒ Logit lens involves directly applying the unembedding matrix to intermediate hidden states to inspect top-ranked tokens.
- ✓ ☐ Logit lens requires training a new linear classifier on top of every intermediate layer.
- ✓ ☐ Logit lens is a type of knock-in experiment.

Done